

9. Vectors

What students need to learn:

Vectors in two and three dimensions.

Magnitude of a vector.

Students should be able to find a unit vector in the direction of $\mathbf{a}$, and be familiar with $|\mathbf{a}|$.

Algebraic operations of vector addition and multiplication by scalars, and their geometrical interpretations.

Position vectors.

$$\overrightarrow{OB} - \overrightarrow{OA} = \overrightarrow{AB} = \mathbf{b} - \mathbf{a}.$$

The distance between two points.

The distance d between two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given by $d^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2$.

Vector equations of lines.

To include the forms $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ and $\mathbf{r} = \mathbf{c} + t(\mathbf{d} - \mathbf{c})$. Intersection, or otherwise, of two lines.

The scalar product. Its use for calculating the angle between two lines.

Students should know that for

$$\overrightarrow{OA} = \mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \text{ and}$$

$$\overrightarrow{OB} = \mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k} \text{ then}$$

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3 \text{ and}$$

$$\cos \angle AOB = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}.$$

Students should know that if $\mathbf{a} \cdot \mathbf{b} = 0$, and $\mathbf{a}$ and $\mathbf{b}$ are non-zero vectors, then $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.
